

KIT CARSON COUNTY AP, CO, USA

WMO: 724689

Lat: 39.245N

Lon: 102.284W

Elev: 1278

StdP: 86.89

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 03026

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.4	-14.4	-21.2	0.7	-14.9	-18.6	0.8	-10.0	15.4	0.2	13.6	1.7	4.7	340	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	36.3	18.3	34.7	18.1	33.0	18.0	20.9	29.4	20.3	29.1	19.7	28.7	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	15.8	23.5	17.8	14.9	22.7	17.2	14.4	22.3	66.4	29.4	63.9	28.9	61.7	28.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.0	10.8	-22.2	38.6	2.7	1.3	-24.1	39.5	-25.6	40.3	-27.1	41.0	-29.1	41.9		
(5)			-22.5	22.3	2.6	0.9	-24.4	22.9	-25.9	23.4	-27.4	23.9	-29.3	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-0.9	-0.3	5.1	9.6	14.8	21.3	24.4	22.8	18.4	10.8	4.5	-0.9
	DBStd	10.34	5.87	6.36	5.65	5.09	4.82	3.91	2.81	2.92	4.51	5.43	5.71	5.76
	HDD10.0	1451	337	292	167	67	13	0	0	0	3	56	179	337
	HDD18.3	3244	596	523	409	263	128	17	2	4	54	237	416	595
	CDD10.0	1767	1	2	17	55	163	338	447	397	254	81	12	1
	CDD18.3	517	0	0	0	1	19	105	190	143	56	3	0	0
	CDH23.3	6957	0	3	25	100	429	1421	2353	1624	830	164	8	0
	CDH26.7	3354	0	1	3	20	152	696	1287	791	362	44	0	0
Wind	WSAvg	5.3	5.1	5.5	5.8	6.4	5.9	5.5	4.9	4.7	5.0	5.2	5.2	5.2
Precipitation	PrecAvg	393	8	10	23	34	74	60	55	48	33	21	13	10
	PrecMax	706	45	70	124	86	117	200	163	164	114	76	40	37
	PrecMin	216	0	0	1	0	9	8	13	0	0	0	0	0
	PrecStd	103	9	14	28	21	31	41	34	38	30	20	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.1	26.4	29.0	33.0	38.1	37.9	37.0	35.2	30.4	24.6	17.9
		MCWB	6.6	8.0	10.5	11.9	15.7	17.2	18.4	18.3	16.9	13.5	9.9	6.9
	2%	DB	15.9	17.3	22.9	26.3	30.7	35.3	36.6	34.7	32.8	27.7	21.8	14.9
		MCWB	5.6	6.4	9.0	11.2	14.8	17.4	18.5	18.2	16.5	12.7	9.0	5.3
	5%	DB	12.1	13.8	19.7	23.6	27.9	33.2	35.2	33.0	31.1	24.9	18.1	11.8
		MCWB	4.1	4.8	8.3	10.8	14.6	17.2	18.6	18.3	16.2	12.0	7.8	3.9
	10%	DB	8.6	10.2	16.3	20.6	25.6	30.9	33.5	31.4	28.4	21.5	14.8	8.5
		MCWB	2.5	3.2	7.0	10.3	13.9	17.3	18.8	18.6	15.8	10.9	6.3	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	8.9	12.4	14.4	18.4	21.0	22.1	21.6	19.6	16.0	11.2	7.4
		MCDB	17.3	20.3	21.9	22.8	27.6	29.3	30.1	29.6	28.6	24.2	20.5	16.8
	2%	WB	5.7	6.7	10.3	13.0	17.0	19.9	21.1	20.7	18.5	14.4	9.8	5.7
		MCDB	14.6	16.2	19.7	21.5	25.5	28.6	29.6	29.2	27.3	22.5	19.7	14.0
	5%	WB	4.3	5.2	8.9	11.8	15.9	19.0	20.5	20.2	17.8	13.0	8.4	4.1
		MCDB	11.5	13.0	18.6	20.4	24.6	28.2	29.4	28.7	26.6	21.9	17.6	11.2
	10%	WB	2.6	3.5	7.3	10.6	14.9	18.2	20.0	19.5	16.8	11.7	6.8	2.6
		MCDB	8.4	10.0	15.6	19.4	23.0	27.5	29.2	28.0	25.1	19.9	14.4	8.3
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.3	15.9	15.6	15.4	15.7	15.6	14.8	15.8	15.7	15.6	14.5
		MCDBR	19.6	20.2	21.6	21.3	19.5	19.0	18.7	17.7	19.3	20.9	20.8	19.0
		MCWBR	11.2	11.2	10.6	9.5	7.7	5.8	5.3	5.4	6.3	9.0	10.5	11.2
	5% WB	MCDBR	18.4	19.0	19.7	17.8	16.8	16.4	15.5	14.8	16.0	17.5	19.7	18.0
		MCWBR	11.0	11.0	10.2	9.1	7.5	6.3	5.5	5.5	6.0	8.5	10.2	11.0
Clear-Sky Solar Irradiance	taub	0.245	0.254	0.280	0.303	0.329	0.348	0.364	0.352	0.321	0.283	0.255	0.246	
	taud	2.580	2.571	2.500	2.460	2.448	2.422	2.426	2.437	2.503	2.578	2.581	2.566	
	Ebn at Noon	958	990	986	974	948	926	907	913	927	940	936	930	
	Edh at Noon	69	81	97	108	112	115	114	109	96	79	69	65	
All-Sky Solar Radiation	RadAvg	2.58	3.39	4.70	5.76	6.65	7.45	7.38	6.50	5.32	3.87	2.83	2.26	
	RadStd	0.18	0.27	0.37	0.28	0.40	0.31	0.27	0.27	0.28	0.32	0.20	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page